Centaur: Whole-Body Loco-Manipulation and Morphology-Control Co-Design

RESEARCH PROPOSAL • UNSTRUCTURED & NON-STATIONARY ENVIRONMENTS

Executive Summary: Legged robots in harsh terrain must move and manipulate as one, yet body and controller are typically designed separately, and a single arm limits work capacity. *Expected results:* Centaur, a six-limbed quadruped that rears to a biped for two-arm work, featuring whole-body RL control and co-designed morphology. *Originality:* Reconfiguration is treated as a learned decision, and the mechanism is co-designed with the policy, built on a working single-arm bipedal-standing prototype.

Problem & Motivation

The world is changing fast, marked increasingly by natural disaster, war, and climate extremes. The recent Venezuela earthquakes killed roughly 2,600 people and left thousands trapped as rescuers raced a closing window. Conflict across Eastern Europe and the Middle East has made whole regions hazardous to enter. In the June 2026 European heatwave, public infrastructure buckled, with roads deforming and rail networks failing across the continent. Such events recur, and the broken terrain they leave behind grows more common.

In these settings a robot must do two things at once: traverse ground that defeats wheels and act on it, whether to clear debris, deliver a payload, or place a sensor near a trapped person. My conviction is that robots should extend human capability rather than replace it, and the sharpest form of that idea is a machine that reaches where a human should not, while a human stays in authority over what it does.

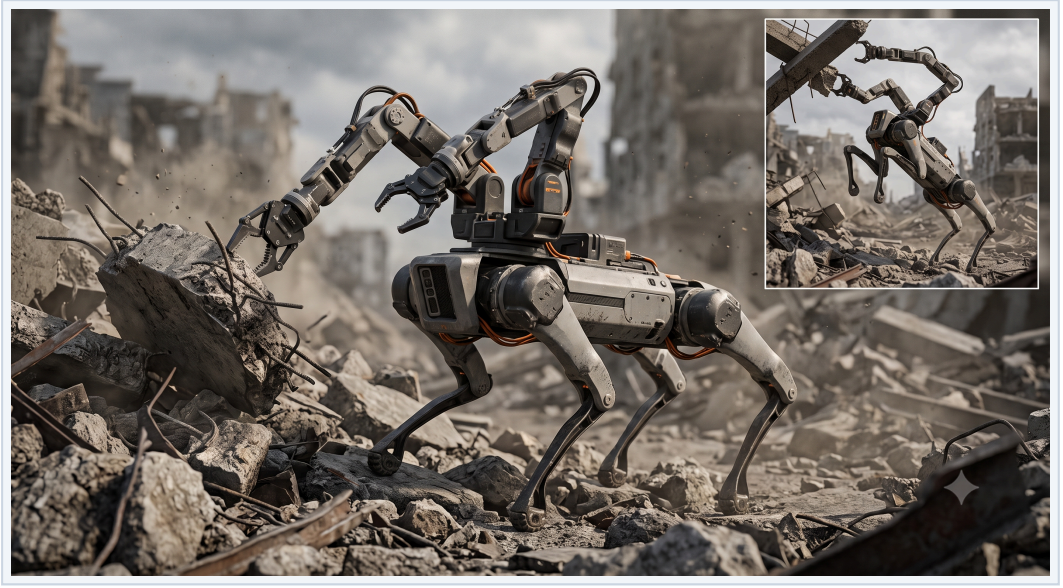


Figure 1: The proposed Centaur platform operating bimanually in non-stationary, post-disaster terrain.

State of the Art & Limitations

Whole-body loco-manipulation treats locomotion and manipulation as one coupled problem, the legs acting as the arm's outer joints, rather than a mobile base with a bolted-on arm. Three limitations remain open and compound in harsh environments. The first is capability. A single arm allows only one-handed tasks, whereas debris clearing, bracing, and carrying need two. The second is embodiment. A fixed quadruped stance commits all four limbs to support, so freeing limbs for manipulation requires reconfiguring the body. The third, and most fundamental, is that mechanism and controller are designed in sequence, the body fixed first and a controller learned for it, which forecloses designs that only a jointly optimized body and brain would find. Co-designing morphology and control is recognized but hard, because each candidate body normally demands a freshly trained controller.

Evidence of Feasibility

In about fifteen hours of GPU time on a single laptop, I built a six-phase skill stack on a Unitree Go2 fitted with an SO-101 arm (18 actuated degrees of freedom). I reproduced the core of a 2025 whole-body loco-manipulation method on different hardware and simulator, substituting a procedural min-jerk trajectory library for its real-world dataset while keeping its adaptive task sampling, reward-factor curricula, and trajectory-velocity-prediction network intact. This is a reproduction with substantial engineering substitutions.



The controller reaches 1.3 to 1.8 cm end-effector tracking, holds 1.4 cm mean error under live teleoperation from a physical leader arm on unseen motion, and keeps perception entirely out of the control loop. I then made the same arm-carrying platform rear onto its hind legs and walk bipedally, balancing at a 0.7 degree mean pendulum angle, adapting a published bipedal-quadruped method to an arm-carrying platform. Reaching a walking bipedal policy took five diagnosed reward iterations.

Proposed Research: The Centaur Platform

Centaur is a legged platform whose name fits its form. Like the mythical centaur, it has six limbs on one body, four legs dedicated purely to locomotion over terrain that stops wheels, and two arms mounted on the trunk for manipulation. In its default quadrupedal stance all four legs support and move the body while the two arms work. When a task demands more reach, it rears onto its hind legs, so the two arms operate at height in a bimanual, human-like configuration. The original contribution is to treat reconfiguration as a learned decision and to co-design the body with the policy, across three parts.

Part 1: Dual-Arm Whole-Body Control

I will extend one-policy becoming a paired-trajectory library with relative-pose constraints between hands, and the adaptive sampler balancing per-pair difficulty. The harder problems I anticipate are reward coupling, support-polygon management under bimanual loads, and the mount redesign for a second arm.

Part 2: Morphological Reconfiguration

My phases already produce distinct policies for distinct stances, so the question is not whether the robot can stand, which it can, but when and how it should choose to reconfigure. Quadrupedal stance buys stability and speed, bipedal stance buys workspace and a free limb pair at the cost of a far smaller support region. I will treat stance as a latent mode with a learned mode-selection layer trained on task outcomes, and with the rear-up and sit-down maneuvers trained as first-class skills that inherit fall recovery from a no-termination design.

Part 3: Co-Design of Mechanism and Control

Rather than fixing Centaur's body and then learning a controller, I want to optimize morphology and policy together, so that limb proportions, arm placement, and actuators are selected for the tasks at hand. I bring a decade of manufacturing experience to understand what is fabricable. One finding already points the way: I learned on the biped that reward frames are part of morphology, because body-frame conventions inherited from locomotion silently assume a horizontal body and break when the body reconfigures. That coupling of body and objective is co-design in miniature.

Outer Directions & Future Integration

Through the clean trajectory interface the platform exposes, a vision-language-action layer decides where the hands go while the controller owns balance. World models enter in ranked order with residual dynamics from real logs highest-value before hardware. Furthest out, a Centaur should pass on what it learns through policy merging, letting robots co-acquire skills without centralizing raw data.

Relevance to Broader Themes

The same platform and control stack serve broader research themes. A humanoid companion for the old or the young must be physically useful, not only emotionally present, able to fetch, steady a person, and act at human height, which is exactly whole-body control with the reconfigurable, upright, arm-carrying body my bipedal-standing result already prototypes. For SLAM on legged robots in changing environments, my controller's deliberate separation of perception from the balance loop is the enabling property, because a mapping and localization module can serve the task-space interface without destabilizing balance. LiDAR mapping with Cartographer, extended-Kalman fusion of odometry and inertial data, and particle-filter localization with scan-to-map matching are competences I already have on wheeled platforms.

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